Meeting Summaries of March 26

Date: March 26
Time: 18:00 - 18:50 PM(PDT)
Attendees: Prof. Jan, Dr. Yue, Jian and Feng
Topic: Discussion, feedbacks and plans about Feng's work on Chapter 2; Clarification to the **logical fallacies** in the use of canonical clustering tool and uncanonical clustering target.

Questions, Feedbacks and Suggestions

- Prof. Jan: Before modeling this MRT or LDA, Feng should get clear insights and pictures about the taxa assemblages, set assumptions with more 'realistic' taxa clusters that the MRT or LDA should target to reveal. But the present work **skips** this 'getting clear insight on taxa assemblages" step, and directly lets the MRT to take over the clustering identification, which uses the five environmental covariates to shape first, then comes to the taxa clusters.

  > Rephrasing this mistake
  >
  > We prioritize the **taxa assemblages first** and want to get clear pictures about the taxa clusters through uncanonical clustering tools that only use the taxa matrix. However, Feng accidentally **used a canonical clustering tool to answer an uncanonical responsible question**. The canonical tool also gives clustering results that are similar to the previous obtains from Ward's hierarchical clustering (uncanonical), so it is a bit confusing and very easy to mess up the logic, the question and its answer.

- Prof. Jan: To make the procedure correct and logical, Feng can follow the **5 objectives** that Prof. Jan has suggested in the email, firstly identifying the **k** 'true' taxa clusters and testing their robustness, explaining their ecological interpretations. Next, carry these **k** clusters to the next step of modeling the relationship between the taxa clusters and the environmental covariates, which gives a classifer that predicts cluster membership with the environmental features. Then, the evaluation of the classifer is another story belonging to machine learning or statistical modeling problems, which will be done with Feng's work.

- Prof. Jan: Specifically, Feng can first use the **Ward's clustering** method to identify the 'true' taxa clusters, then apply the **MRT** method to train a classifer that correlates the environmental covariates with the taxa clusters.

- Dr. Yue: Students are always easily trapped by the technically analytical results and forget their explanability to real wrod meaning. Especially when some analytical tools are complicated and provide **similar results**, students may easily ignore these dissimilarities and passively accept the results without **critical thinking**.

- Jian: Most of the figures are too **dense with many subplots**, not only the figure with 16 taxa error-bar plots, but also the figures with map and barplots of three clusters. It is hard to explain their information in a short presentation to the audience. Maybe Feng could make the figures simpler with a single plot for clearance, sometimes using a table to organize the results is also a good choice.

Action Items

- Feng will **correct and modify** the current analysis procedure for invertebrate assemblage, following to the suggestions that Prof. Jan has given.
- Feng will **stop** analysis and coding work for one week, and do **writing and organizing work** for this week. The chapter - **Contamination Assessment** and the chapter - **Invertebrate Assemblage** have brought a lot of analytical results and critical thinking that need to be organized and written down in a clear way, specifying what we have done and what we did not or cannot.

Summary of Discussion

- Feng gave a report on the results by **Multivariate Regression Tree (MRT)**, it is a canonical clustering tool and produces clusters of response variables while under the control of explanatory variables. Specifically, Feng introduced the **steps** of implementing MRT and the finalized results through the procedure as the book (Numerical Ecology text) recommends:
    1. MRT allows **explanatory variables** to divide the samples first, this division only uses explanatory matrix and does not consider the **response (taxa) matrix**.
    2. After the division with distinct explanatory variables and their thresholds, each divided cluster of samples will be evaluated through the use of **response (taxa) matrix**. Specifically, the evaluation here is the ratio between **within-group sum of squares (WSS)** and the total sum of squares (TSS) in the response variable space. The globally best division (goal) is to minimize the **relative error** (RE) which is the ratio of WSS and TSS, the lowest RE gives the best division.
    3. In the selected 25% sites as least polluted sites, the MRT procedure gives the best division of 3 clusters with its estimates of RE, CVRE, SE of the CVRE (by the permutation test for the estimating process of CVRE). The specific tree structure is shown and the clusters environmental and taxonomic characteristics are shown as well.
    4. Feng thought that the finalized three groups by MRT are the **'true' taxa clusters** (but it is incomplete in evidence through this proving way) in the collected samples. It made a logical fallacies because the MRT is a canonical clustering tool that emphasizes the explanatory variables first, then comes to the response variables. It is **not** a clean tool to reveal the **'true' clusters** of taxa in the samples, but more restricted by the explanatory variables.

Reflection on MRT and its explanatory variables

Canonical clustering tools allow the explanatory variables to shape the clusters first, which can be predictive and useful for figuring out the relaitonship between the response clusters and explanatory variables. The closer these explanatory variables are to the **'true' drivers** of the response clusters, the **more realistic** the clustering results will be, leading to similar results as the 'true' clusters of taxa that we assumbly would obtain from **uncanonical** clustering tools that use only the response variables to cluster the samples.

- However, MRT can be a powerful **predictive** tool to model the non-linear relationship between the environmental covariates and the taxa clusters. It provides a possible way to train a good classifer for prediction of new samples with their environmental covariates.

Follow-Up

Temporary meeting time: April 2, 18:00 - 18:50 PM(PDT)

Supportive materials

- A meeting recording with subtitles has been restored on Google Drive.